ALOK SINGH KATARIA

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DEVOPS · SRE · CLOUD ENGINEERING · SOFTWARE ENGINEERING · AI · QUANTUM COMPUTING

OBJECTIVE

Seeking a senior **DevOps, SRE, or Cloud Engineering** role within IBM where I can deepen my contribution to IBM Cloud platform reliability, compliance automation, and cloud-native infrastructure. With 3+ years of current IBM experience across Kubernetes, OpenShift, IaC, CI/CD, and compliance engineering — and 9 years of prior IBM product engineering experience — I am committed to continued growth and long-term impact within IBM.

PROFILE

Advisory Software Engineer with 25+ years of professional experience, including **12+ years at IBM** across DevOps, SRE, cloud platform engineering, compliance automation, and enterprise software development. **IIT Kanpur Physics graduate** with a strong analytical and mathematical foundation.

Currently at IBM, driving cloud-native platform reliability, Kubernetes-based infrastructure modernization, IaC-based re-platforming, and Python-based compliance automation across IBM Cloud. Hands-on across the full DevOps and SRE stack — Kubernetes, OpenShift, Docker, Jenkins, Tekton, IBM Cloud VPC, IAM, and CI/CD pipelines.

Combines deep systems understanding with strong software engineering skills to **design resilient infrastructure, automate compliance workflows, and reduce operational toil at scale**. Additionally brings hands-on experience with AI/ML tooling and IBM Quantum computing — areas of active IBM investment.

Committed to IBM. Focused on DevOps, SRE, Cloud Engineering, and Platform Reliability — with the breadth and depth to contribute across IBM Cloud and beyond.

CORE AREAS

DevOps & SRE · Kubernetes & OpenShift · IBM Cloud · Cloud-Native Infrastructure
CI/CD Engineering · Compliance Automation · IaC · Platform Resilience
Python Automation · AI-Assisted Engineering · Disaster Recovery
Quantum Computing · Computational Physics · Scientific Computing

PROFESSIONAL EXPERIENCE

IBM — Advisory Software Engineer | DevOps / SRE | 2022 – Present

1. CSUtils Ecosystem — Platform Architecture, Resilience & L3 Engineering

- Led architecture modernization and resilience engineering across IBM Cloud security utilities, modernizing Kubernetes configuration with CRDs, strengthening VPN connectivity with fail-fast validation, and ensuring reliable syslog delivery across pod restarts.
- Designed automated multi-cluster chaos testing and owned L3 production troubleshooting across Kubernetes, VPN, TCP/UDP connectivity, and networking failures to improve service reliability.

2. Dreadnought — IBM Cloud PaaS Re-platforming & Cloud Engineering

- Drove IaC-based provisioning and re-platforming of IBM Cloud infrastructure, including VPC networking, OpenShift, IAM, managed databases, and CI/CD integrations using standardized platform configurations.
- Implemented private endpoint connectivity, PAG/GPVPN integration, bastion access, and security scanning; supported complex cloud networking and platform escalations through root-cause analysis and remediation.

3. Compliance Automation — AI Agent, Auditree & CI/CD Engineering

- Led Auditree onboarding and Python-based compliance automation for SOC 2-aligned evidence collection, security posture assessment, centralized evidence storage in IBM Cloud COS, and Git-based remediation workflows.
- Initiated an AI agent for troubleshooting Jenkins and Auditree pipelines, leveraging repository documentation for failure analysis and remediation recommendations; integrated automated workflows with CI/CD practices spanning Jenkins and Tekton.

4. Armada Compliance Engine — Compliance Automation, SRE & DR Hardening

- **Led Compliance Modernization & Process Improvement:** Spearheaded the transition from legacy, high-latency node-level BigFix agents to a centralized, automated Go-based event-driven compliance engine, drastically reducing inventory lag and automating real-time registration/deregistration at scale.
- **Headed Incident Resolution & Diagnostic Optimization:** Resolved critical, silent compliance-sync defects (SOS API silent failures); designed and published an end-to-end operational runbook with standard decision trees, search signatures, and `@xo` CLI diagnostic flows, reducing mean-time-to-resolution (MTTR) for on-call teams.
- **Engineered DR Resilience & CI/CD Integrity:** Enforced disaster recovery (DR) readiness by configuring zone-spread constraints and high scheduling priorities (`system-cluster-critical`) to prevent node-starvation outages during brownfield migrations, while debugging and resolving broken Jenkins regional carrier lookup and build env pipelines.

Wipro — Lead Software Developer | 2022

Software development for the **CreditMemo** process of the Commissioning System for **Telefónica Germany**, building Python/Flask/Pandas services for data loading, aggregation, and validation.

- Implemented feature enhancements, defect fixes, and unit testing; worked with PostgreSQL and Google Cloud.

RevenueBox — Senior Software Engineer | 2021

B2B go-to-market platform focused on account identification, prioritization, and engagement automation.

- Developed a Python microservice to process and rank customer accounts based on multiple input signals.
- Designed a mathematical scoring model (**"Quibble Score"**) for account prioritization, using Python, Docker, MongoDB, and Google Cloud.

Lumos Learning — Senior Software Engineer | 2019 – 2020

Cloud-native Python microservices and serverless platform development for a high-traffic K-12 learning platform serving millions of users.

- Built **Python-based serverless services** using AWS Lambda, automating content generation, scoring pipelines, and adaptive evaluation workflows at scale.
- Developed NLP-based automation using Python and NLTK for intelligent content classification and question-generation pipelines.
- Architected and delivered the **serverless cloud backend** on AWS (Lambda, SQL Server) — owning end-to-end system design, implementation, and deployment.
- Applied mathematical modelling and algorithmic design to difficulty estimation and adaptive scoring, bringing structured analytical rigour to product engineering.

HCL Technologies — Senior Software Engineer | 2017 – 2018

L3 product support for **IBM AppScan** (web application security testing), troubleshooting complex customer issues, root-cause analysis, and coordinating fixes across SAST/DAST scanning and reporting workflows.

- Worked on AppScan SAST/DAST capabilities, customer configurations, defect analysis, and production issue resolution.

IBM India Software Labs — Senior Staff Software Engineer | 2008 – 2017

L3 Development & Troubleshooting Engineer for **IBM InfoSphere Data Replication (IIDR)** (previously DataMirror), specializing in complex production issues, root-cause analysis, debugging, and customer escalations.

- Expert in heterogeneous data replication and integration, including **DB2 for i / AS400, Oracle, DB2, SQL Server, and Teradata**, with IBM messaging and remote journaling technologies.

- Strong experience in cross-language and cross-platform integration, including **AS400/IBM i** and COBOL-based enterprise systems; supported customer ETL and data-movement workflows.
- Recognized as a performance specialist, contributing to performance analysis and optimization and receiving an **IBM award for performance expertise**.
- Contributed to feature development, product maintenance, automation, and build/release pipeline engineering across multiple product releases.
- Worked with C, Java, Python, RPG, COBOL, AS400, and Jenkins.

CA Technologies — Software Engineer | 2005 – 2008

Worked on **eTrust Access Control**, **eTrust Audit**, and **eTrust Single Sign-On** — enterprise security products.

- Developed more than 100 automated tests for nightly builds and ported product code to AIX.
- Investigated customer and QA defects including memory corruption, enhancements, and installation/upgrade issues.
- Worked with C, Python, Linux, UNIX, AIX, Solaris, and HP-UX.

TECHNICAL SKILLS

Programming: Python, C, Java, Shell, JavaScript

DevOps & Cloud: IBM Cloud, AWS, Google Cloud, Kubernetes, OpenShift, Docker, Jenkins, Tekton, CI/CD, IaC, Auditree

Scientific & Data: NumPy, SciPy, Pandas, MATLAB, scikit-learn, TensorFlow

Quantum Computing: Qiskit, VQE, Quantum Circuits, Quantum Simulation

AI-Assisted Engineering: ChatGPT, IBM Bob, GitHub Copilot (leveraged for codebase analysis, debugging, refactoring, and structured document engineering)

Databases: PostgreSQL, MongoDB, SQL Server, Oracle, DB2, Teradata

Operating Systems: Linux, AIX, Solaris, HP-UX, Windows, IBM i / AS400

SELECTED RESEARCH & TECHNICAL WORK

Quantum Intuition: From Spin Dynamics to the Hadamard Gate and π Estimation via Quantum Randomness

Explored quantum randomness and quantum-state transformations through Qiskit-based simulations, including spin dynamics, Hadamard transformations, measurement probabilities, and Monte Carlo estimation of π.

Status: Under IBM internal review for publication.

Classical VQE Demonstration for the Hydrogen Atom

Developed a classical demonstration of the VQE approach for the hydrogen atom, connecting the variational principle, Hamiltonian formulation, parameterized states, energy expectation values, and numerical optimization.

Status: Under IBM internal review for publication.

INTELLECTUAL PROPERTY

- **US Patent #9,528,562** — Granted US patent related to mechanics.
- Two additional mechanics-related inventions under IBM patent review.

EDUCATION

IIT Kanpur — Integrated M.Sc. in Physics | 2000

ACADEMIC DISTINCTIONS & AWARDS

- **All India IIT-JEE Advanced Rank** — 1211 (General)
- **GATE All India Rank** — 29 (General) | **Score** — 98.73 (General)
- **IBM Excellence and Eminence Award (2012)** — Received for outstanding L3 support, troubleshooting expertise, and high-priority resolution of complex client issues.

ADDITIONAL BACKGROUND

Physics Teaching & Mentoring (2001–2004, 2017–2018): Taught Physics for IIT-JEE Advanced and NEET at leading coaching institutions across Andhra Pradesh, contributing to 40+ IIT-JEE selections. Designed original IIT-JEE Advanced-level problems and contributed to faculty development initiatives.

RESEARCH & CODE | COMPUTATIONAL AND QUANTUM PHYSICS

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Independent computational physics projects: numerical solutions of the Schrödinger equation, Hamiltonian modelling, quantum circuit simulation, VQE, quantum chemistry, quantum optimization, and quantum-information experiments.